Sam Stowers

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PROFESSIONAL SUMMARY

MS Computer Science graduate from Syracuse University (4.0 GPA) completing a combined BS/MS in 4 years, with published-track research in Linux kernel KVM optimization and hands-on systems programming in C and C++. Proven ability to diagnose low-level performance bottlenecks and ship kernel-level solutions in multi-core virtualized environments.

EDUCATION

Syracuse University — Syracuse, NY
Master of Science in Computer Science — May 2026
GPA: 4.0/4.0 **Track:** Accelerated BS/MS (4-year track)
Thesis: "SAGE: Contention-Aware Sensitivity Adjustment for Spin Lock Exiting in VMs" (Advisor: Prof. Bryan Kim)

Syracuse University — Syracuse, NY
Bachelor of Science in Computer Science — May 2025
GPA: 3.99/4.0 **Honors:** Summa Cum Laude, Tau Beta Pi, Chancellor's Scholar
Minors: Mathematics, Logic

Bellarmine College Prep — San Jose, CA
High School Diploma — May 2022

PROJECTS

Linux KVM Optimization Research – *C, Python, Bash, Linux Kernel* (August 2025 - May 2026)
- Replicated cloud virtual machine instances using Docker images on production-grade systems to quantify performance degradation from neighbor interference
- Designed and implemented SAGE (Sensitivity Adjusted Gap Exiting) within the Linux kernel KVM module, introducing a dynamic tuning mechanism that achieved a over 2× speedup under high vCPU contention
- Authored a comprehensive technical manuscript targeted for a peer-reviewed systems conference and thesis defense

Predictive Air Quality AI Model – *Python, XGBoost* (April 2026)
- Designed end-to-end XGBoost machine learning pipeline utilizing NOAA GFS data to predict city-level air quality with 14% mean error for a 72-hour horizon
- Awarded for most accurate model out of 40 graduate students

Code Bounty Board – *C++, Ethereum, Flask* (March 2026)
- Architected a decentralized platform with C++ validation suite, ETH payment channels, and Ethereum Smart Contracts

City Sync – *Javascript, React, Firebase* (January - May 2025)
- Developed a platform with user authentication, Google Maps integration, and real-time Firebase databases via Agile/SCRUM

WORK EXPERIENCE

Data Annotation | Remote
Programmer & Mathematician (September 2024 – Present)
- Conducted code review for hundreds of responses from AI reasoning models, identifying failure modes and capability limitations

Physics Department at Syracuse University | Syracuse, NY
Teaching Assistant (August 2023 – May 2024)
- Conducted and led over 25 different mechanical physics labs alongside a team of graduate students and professors

SKILLS

- **Infrastructure:** Linux, Docker, Kernel Development, KVM Virtualization
- **Languages:** C, C++ (pthreads, OpenMP), Python, Racket, Bash, LaTeX, Java, Javascript, Haskell
- **Tools:** Git, `perf`, GCC, CMake, GDB, GTest, PyTest
- **Frameworks & Libraries:** Pandas, NumPy, SciPy, XGBoost, Flask, React

COURSEWORK

- **Computer Systems:** Advanced Computer Architecture, Principals of Operating Systems, Compiler Design
- **Software Engineering:** Algorithms, Machine Learning, Software Specification and Design, Artificial Intelligence
- **Mathematics and Theory:** Automata and Computability, Formal Methods, Numerical Methods, Applied Linear Algebra